

Parameter	Value	Note
cell birth volume (μm^3)	1	
doubling time (min)	150	Growth condition is ABT minimal glycerol for wildtype MG1655. This is to guarantee non-overlapping cell cycle, which avoids synchronization issue.
C period (min)	70	
D period (min)	30	
replication speed (kbp/min)	66.31	
dnaA transcription rate (mRNA/min) (per gene)	1.5	We simply assume constitutive gene expression at this first stage.
DnaA translation efficiency (protein/mRNA)	1	Estimated by Hansen & Atlung 2018.
DnaA degradation rate	0	assuming it is fully stable
DnaA to ATP dissociation constant (nM)	30	Measured by Sekimizu et al. 1987 and Kawakami et al. 2006.
DnaA to ADP dissociation constant (nM)	100	
DnaA-ATP intrinsic hydrolysis rate (/min)	0.046	Measured by Sekimizu et al. 1987
consensus DnaA box locations	(see the document linked on the right)	dnaa_box_loci_locations Sheet "DnaA boxes"
consensus DnaA box dissociation constant (nM)	1	ignore their differences first; same for either DnaA-ATP or DnaA-ADP
oriC location	3925744-3925989	dnaa_box_loci_locations Sheet "locus locations"
oriC low-affinity DnaA-ATP box number	8	binds DnaA-ATP only
oriC low-affinity DnaA-ATP box dissociation constant (nM)	50-1	Considering cooperativity, the dissociation constant decreases with occupancy, from 50 nM (first binding site) to 1 nM (last binding site), assumably linearly.
Hda hydrolysis rate (/min) (per replication fork pair)	40	artificial setup
datA location	4392732-4392914	dnaa_box_loci_locations Sheet "locus locations"
DnaA-ATP hydrolysis rate at datA (/min) (per datA locus)	12	artificial setup
DARS1 location	813086-813186	dnaa_box_loci_locations Sheet "locus locations"
DnaA-ADP to DnaA-ATP conversion rate at DARS1 (/min) (per DARS1 locus)	5	artificial setup

Parameter	Value	Note
DARS2 location	2969112-2969367	dnaa_box_loci_locations Sheet "locus locations"
DnaA-ADP to DnaA-ATP conversion rate at DARS2 (/min) (per DARS2 locus)	10	artificial setup
SeqA sequestration time (min)	0	not considered in the beginning